

Jorge Rodríguez Ruigómez

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Engineer focused on Applied AI and Forward Deployed Engineering, with experience in production AI automation, multimodal retrieval and user-facing software. M.S. student at Illinois Tech.

Work Experience

AI Research Assistant | Universidad Politécnica de Madrid Oct 2025 – Aug 2026
Madrid, Spain

- Built an end-to-end multimodal AI system for MOTIVATE-XR, transforming video into searchable knowledge for grounded question answering with vision-language models, speech recognition and LLMs.
- Designed and evaluated RAG pipelines combining vector databases, knowledge graphs and intent-aware hybrid retrieval; enabled multi-step agent reasoning through MCP-based tools.
- Designed the chatbot system architecture and containerized the system with Docker for reproducible deployment and experimentation.

AI Engineer Intern | NTT DATA Feb 2025 – Jul 2025
Madrid, Spain

- Contributed to production banking document automation for Banco Sabadell using generative AI and Intelligent Document Processing, including inheritance deeds and tax settlements.
- Developed and iterated on LLM prompts for structured extraction; built Python preprocessing and output-validation functions to standardize data and support downstream workflows.

Product Development & Leadership

Founding Team Member | ETSIT-UPM Alumni Network Oct 2025 – Present
Software development & product delivery | Remote

- Helped build and launch the platform in July 2026 as part of a five-person team, working directly with the School's leadership on product design, development and launch.
- Developed registration, QR membership credentials, admin validation and event-access workflows; integrated payment and QR-generation APIs and continue to maintain and improve the platform.

TelecoEmprende | ETSIT-UPM entrepreneurship club Oct 2025 – Present
Board Member (Aug 2026–Present); Vice President (Oct 2025–Aug 2026)

- Helped relaunch the club and grow its active community to 80 members; co-organized three events with 100+ attendees featuring founders, executives and investors.

Education

Illinois Institute of Technology Aug 2026 – May 2027
M.S. in Information Technology and Management | Applied Data Science & AI

Universidad Politécnica de Madrid Sep 2025 – May 2027
Master's in Telecommunications Engineering | Machine Learning & Data Science

Universidad Politécnica de Madrid Sep 2020 – Jun 2025
B.Eng. in Telecommunication Technologies and Services

- **Thesis: Intelligent routing for generative AI models**, optimizing accuracy, cost and consumption. Grade: 10/10; nominated for an honors distinction.

Technical Skills

Programming & data: Python, JavaScript, SQL, Java, C; PyTorch, Pandas, NumPy

Applied AI: LLMs, AI agents, RAG, multimodal AI, MCP, knowledge graphs, prompt engineering, IDP

Engineering: API development & integration, Git, Docker, Kubernetes, Azure AI, data validation

English: C1 (Cambridge Advanced); TOEFL iBT 104/120

Awards

Repsol Challenge Winner – IndesIAHack Nov 2024
Team project: AI assistant, energy-savings calculator and automated social content workflow.

2nd Place – Gaia-X Hub Spain Hackathon Dec 2024
Team project: Data Space Connector ETL pipeline and AI-based housing location assessment prototype.